

Fig. 8: Flow-chart of keyboard subroutine

matches with the value of the 5x4 matrix of the matrix array 'key_pad [5][4]' declared in the program.

Row reading.

1. Starting with the top row, the microcontroller makes each row high at a time and then reads the columns. If the data read is all zeros, no key in that row is activated and the program moves to the next row.

2. It makes the next row high, reads

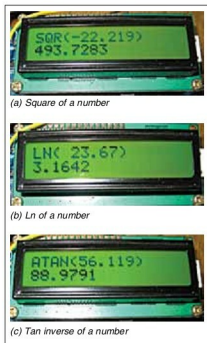


Fig. 9: User manual of the calculator for square, natural log (Ln) and tan inverse

the columns and checks for any high.

3. This process continues until the row having a key pressed is identified, i.e., reading of the column returns a non-zero value.

4. After identification of the row in which the key has been pressed, the program finds out which column the pressed key belongs to.

Column reading. The microcontroller reads the data at Port A and masks its lower bit.

Variables 'row' and 'column' are global, and they can be used from anywhere in the program. For reading a switch which has more than one operation, timer1 is started before entering the function 'keyboard()'. The program goes on reading the keyboard until timer1 stops counting. When timer1 finishes counting, it returns from the function.

Key multiplexing is done with the help of the timer. At first, the timer is initialised to 16 bits and 'TMR0IF' is set to 0. The timer starts counting only when a value is written to it. After it finishes counting, 'TMR0IF' becomes 1. This timer is used only for those keys which perform mathematical operation.

eration. For keys other than operator keys, it does not start counting. But at the press of any operator key, 'Write-Timer0(50000)' is used to write value to the timer and the timer starts counting. If the same key is pressed again before the timer completes counting, the second value of the key gets accepted and the timer value is again set to 50000. The process goes on. This way, key multiplexing is done and a single key can perform more than one operation.

'Read from LCD' subroutine. This subroutine reads numbers from the LCD and returns them as signed, double-data type. The cursor goes to the LCD location (given by the user) from where the data is to be read. The program reads the number, sign and decimal point from the LCD one by one, and stores them in string array 'ch[i]'. Then it converts the string array into signed, double-data type 'ans' using function 'atof()' and returns the value of 'ans.' Now the microcontroller has the number that was typed into the LCD and any mathematical operation can be performed on it.

'Write result to LCD' subroutine.

This subroutine writes the result to the LCD. It receives the LCD location where the data is to be written, the signed floating point number and the number of places to be displayed after the decimal point.

This function also displays word 'Error' on the LCD if the result of any mathematical operation is out of range or mathematically incorrect.

Input number subroutine. This function takes operand inputs for mathematical calculations. For mathematical operations requiring two operands (like addition and subtraction), it is executed twice. For mathematical operations requiring a single operand (like log), it is executed only once. It first takes the input of the sign bit, then the integer part followed by the decimal part.

The function uses time multiplexing of the sign (+/-) switch with the